Multiple Probabilities of Direction: A Rationale for Prior-odds Adjustment

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Abstract

In psychological research, Bayesian analyses are often summarized using posterior indices and then interpreted with conventional cutoffs across many parameters. The probability of direction (*pd*) is one such index, defined as the posterior probability that a parameter is greater or smaller than a null value. When *pd* is thresholded and used repeatedly, multiplicity becomes consequential but is rarely handled explicitly. Here, we examine the operating characteristics of *pd* under repeated use by leveraging its correspondence with the frequentist *p*-value under common settings. Simulations show that extreme *pd* can arise by chance and that misleading conclusions become more frequent as the number of tests increases. We further express *pd* in terms of posterior odds and propose post-hoc prior-odds adjustments that provide multiplicity control. To support practical implementation, we provide an R package that applies the proposed prior-odds corrections.

Keywords: probability of direction, multiple testing, prior-odds

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Over the past two decades, the proportion of articles reporting Bayesian analyses in psychology has increased considerably (Jevremov & Pajić, 2024; Pfadt et al., 2025; Van De Schoot et al., 2017). Unfortunately, this expansion has been accompanied by practices reminiscent of the "null ritual" in frequentist statistics, where widespread use was associated with automatic application rather than principled reasoning (Cohen, 1994; Gigerenzer, 2004). For instance, researchers sometimes treat Bayes factor thresholds (e.g., BF > 3) as dichotomous decision rules or claim that an effect "exists" because the posterior probability of a positive effect exceeds a conventional benchmark such as 95% (Heck et al., 2023; Hoijtink et al., 2016; Kelter, 2023; Kruschke, 2018; Tendeiro et al., 2024; Tendeiro & Kiers, 2019). While such summaries are convenient, their use as default cutoffs encourages all-or-none interpretations. When these binary judgments are applied repeatedly across many parameters, the issue of multiplicity arises.

In the frequentist framework, multiplicity is typically framed as Type I error (false-positive) problem: if each individual test is conducted at level α, the probability of at least one false positive increases with the number of tests. Classical approaches therefore control either the familywise error rate (the probability of at least one Type I error) or the false discovery rate (the expected proportion of false positives among rejected hypotheses) [**REFs**]. However, multiplicity concerns extend beyond false positives. When researchers examine many parameters and highlight the most extreme estimates, selection effects can produce exaggerated magnitudes (Altoè et al., 2020; Gelman & Tuerlinckx, 2000; Zwet & Cator, 2021). Bayesian shrinkage methods address these distortions by pulling extreme estimates toward more plausible values. Such shrinkage can arise through hierarchical models with partial pooling (Berry & Hochberg, 1999; Gelman et al., 2012, 2013; Kruschke, 2011) and/or through informative priors that reduce overestimation in low signal-to-noise settings (Dawid, 1994; Zwet & Gelman, 2022).

Regularization improves estimation, but it does not by itself calibrate repeated threshold-based decisions across many parameters. If the same cutoff is applied to many parameters (e.g., declaring an effect whenever posterior probability exceeds 0.95), the chance of at

least one spurious finding can remain inflated, even if shrinkage reduces it. Bayesian approaches that directly target this issue typically rely on either simultaneous credible statements (Box & Tiao, 1992; Gelman & Tuerlinckx, 2000; Pruzek, 2016; Schnell et al., 2020) or priors on the hypothesis/model space that scale with the number of tested effects (Jeffreys, 1938; Scott & Berger, 2006; Scott & Berger, 2010; Westfall et al., 1997). Nevertheless, psychological researchers often conflate selection effects that distort estimates with the distinct problem of making many simultaneous decisions. As a result, shrinkage is sometimes treated as sufficient for multiplicity control (Chen et al., 2020; Erčulj & Štrumbelj, 2015; Hogeveen et al., 2022; Limbachia et al., 2021; Mathar et al., 2022).

Other justifications invokes the *subjectivist* interpretation of Bayesian inference, which treats posterior probabilities as rational degrees of personal belief (De Finetti, 2017; Edwards et al., 1963; Lindley, 1972; Rozeboom, 1960). However, this interpretation fits poorly with common psychological practice, where researchers typically apply default software priors (often weakly informative) rather than priors encoding substantive domain knowledge (Smid & Winter, 2020; Van De Schoot et al., 2017). When these priors are combined with sufficient data, Bayesian and frequentist procedures often converge to similar conclusions (Albers et al., 2018; Bernardo & Smith, 1994; Gelman et al., 2013). This convergence makes it both natural and useful to evaluate the frequentist operating characteristics of Bayesian decision rules under multiplicity.

Accordingly, we adopt a *falsificationist* perspective, which treats Bayesian procedures as tools to be calibrated rather than as solely expressions of belief (Box, 1980; Gelman & Hennig, 2017; Gelman & Shalizi, 2013; Mayo, 2018; Rubin, 1984). Here, we focus on the Probability of Direction (*pd*), a Bayesian index that can be mapped directly to the frequentist *p*-value in common settings (Berger & Sellke, 1987; Casella & Berger, 1987; Makowski et al., 2019; Marsman & Wagenmakers, 2017). This correspondence allows us to evaluate *pd*'s behavior under repeated use and to compare Bayesian and frequentist approaches to multiplicity control on common ground.

The paper proceeds in four steps. First, we introduce *pd* and make explicit its connection to the frequentist *p*-value, which provides a reference point for calibration. Second, we

characterize *pd*'s null distribution and examine its multiple-testing behavior through simulations that quantify false-positive rates. Third, we relate *pd* to directional Bayes factors and formalize the role of prior odds. Finally, we present prior-odds adjustments for multiplicity following Jeffreys (Jeffreys, 1938; see also Westfall et al., 1997), discuss implementation under dependence, and provide guidance for reporting both raw and multiplicity-adjusted *pd*.

Probability of direction and p-value

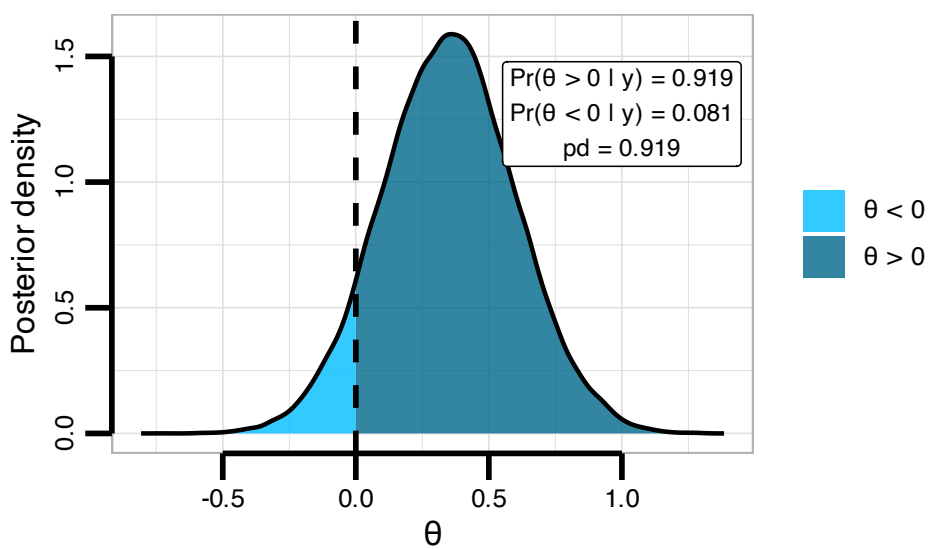
The probability of direction (*pd*) for continuous distributions can be defined as the maximum between the posterior probabilities that a parameter is greater or smaller than the null value:

$$pd = \max(\Pr(\theta < \theta_{null} \mid y),\ \Pr(\theta > \theta_{null} \mid y)).$$

As proposed and popularized in applied Bayesian workflows, *pd* offers an intuitive summary of directional evidence by quantifying the proportion of posterior that lies on the dominant side (Makowski et al., 2019); it ranges from 0.50 (no directional preference) to 1 (all on one side).

Figure 1

Posterior distribution with directional posterior mass on either side of 0 shown in different colors; the probability of direction (pd) is the larger of the two posterior probabilities.



Because *pd* is a functional of the posterior, it inherits dependence on the full model

specification (likelihood and prior), so its long-run error behavior is not a property of *pd* "in isolation" but of the overall Bayesian procedure that generates the posterior.

A convenient starting point for calibration is the mapping between *pd* and the frequentist *p*-value. Under flat (or sufficiently diffuse symmetric) priors in typical hypothesis-testing scenarios (e.g., testing means), *pd* is simply the posterior on the dominant side of θ_{null}, which corresponds to the complement of the one-sided *p*-value (Casella & Berger, 1987; Marsman & Wagenmakers, 2017):

$$pd = \max(Pr(\theta < \theta_{null} \mid y), Pr(\theta > \theta_{null} \mid y)) = 1 - p,$$

with the two-sided correspondence being $p_{two-sided} = 2p = 2(1 - pd)$.

Probability of direction as a test of existence and direction

The *pd* can be embedded in a testing framework, where thresholding $pd > c$ yields a claim that an effect exists and has a particular sign. For example, when $pd > c$ supports $\theta > 0$, the posterior probability assigned to nonpositive values is bounded by

$$\Pr(\theta \leq 0 \mid y) < 1 - c,$$

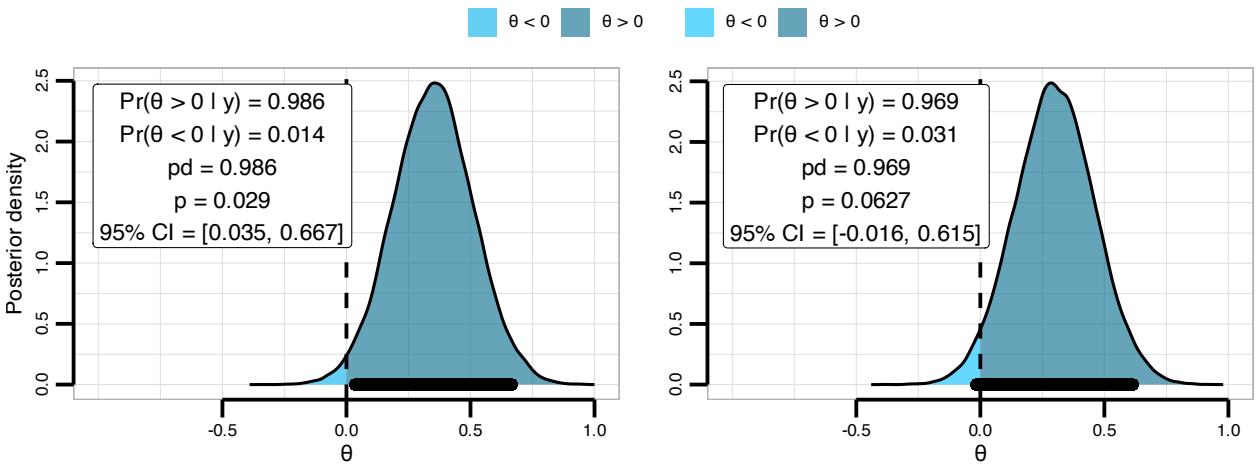
and, analogously, when $pd > c$ supports $\theta < 0$, it follows that $\Pr(\theta \geq 0 \mid y) < 1 - c$.

This rule therefore does more than compare H_+ versus H_-: it also controls posterior mass in the null-and-opposite-sign region, ensuring that it remains below $1 - c$. In particular, $\Pr(\theta \leq 0 \mid y) < 1 - c$ implies that 0 lies below the $1 - c$-quantile of the posterior distribution (equivalently, the $1 - c$-quantile is strictly positive). Consequently, the central credible interval bounded by the $1 - c$ and c quantiles excludes 0.

Finally, c can be set so that the interval between the $1 - c$ and c posterior quantiles coincides with the usual equal-tailed $1 - \alpha$ central credible interval, whose endpoints are the $\alpha/2$ and $1 - \alpha/2$ posterior quantiles. Equating endpoints gives $1 - c = \alpha/2$ and $c = 1 - \alpha/2$; therefore $pd > c$ can be read as excluding 0 from a two-sided $100(1 - \alpha)\%$ central interval. With $\alpha = 0.05$, this yields $c = 0.975$ (and $1 - c = 0.025$), mirroring the common two-sided 5% convention (Figure 2).

Figure 2

Posterior distributions illustrating the relationship between the probability of direction (pd) and credible-interval (CI), using $c = 0.975$ and $\theta_{null} = 0$.



Pure directional testing conditions on the assumption that $\theta \neq 0$ and evaluates only the sign of the effect by comparing H_+ versus H_-; $H_0 : \theta = 0$ is not among the hypotheses under consideration. In contrast, here we explicitly include H_0 and separates decisions about existence and direction (excluding zero, e.g. H_+) from decisions about direction (including zero, e.g. H_{-0}; Goeman et al. (2010)).

Importantly, although we set $H_0 : \theta = \theta_{null} = 0$ throughout the manuscript for simplicity, θ_{null} can be defined as the smallest effect size of interest (Lakens et al., 2018). The testing framework thus shifts from evaluating whether an effect differs from zero to whether it exceeds a substantively meaningful threshold.

Simulation

In our simulations we generate data under the global null (all $\theta = 0$) and we compute $\Pr(pd > c \mid \theta = 0)$. Operationally, this quantifies how often the procedure falsely excludes zero from the credible region (bounded by $1 - c$ and c), that is, incorrectly concludes that an effect exists (with a particular sign). We conduct a Monte Carlo simulation in which, for each condition, we generated families of size $m \in \{1, 2, 5, 10, 20\}$, with $n \in \{30, 50, 100\}$ observations per dataset, and repeated the procedure 10,000 times. In every replication, data were generated under

the global null: each dataset had true mean 0 and unit variance, $y \sim \mathcal{N}(0, 1)$. Each dataset was analyzed using a conjugate normal model with known variance $\sigma^2 = 1$ and prior $\theta \sim \mathcal{N}(0, \tau^2)$, where $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$ spans skeptical to flat priors. We drew posterior samples of θ from the resulting normal posterior and computed *pd* as $pd = \max(\Pr(\theta > 0 \mid y), \Pr(\theta < 0 \mid y))$ using these draws. For the frequentist comparison, we conducted two-sided one-sample z-tests.

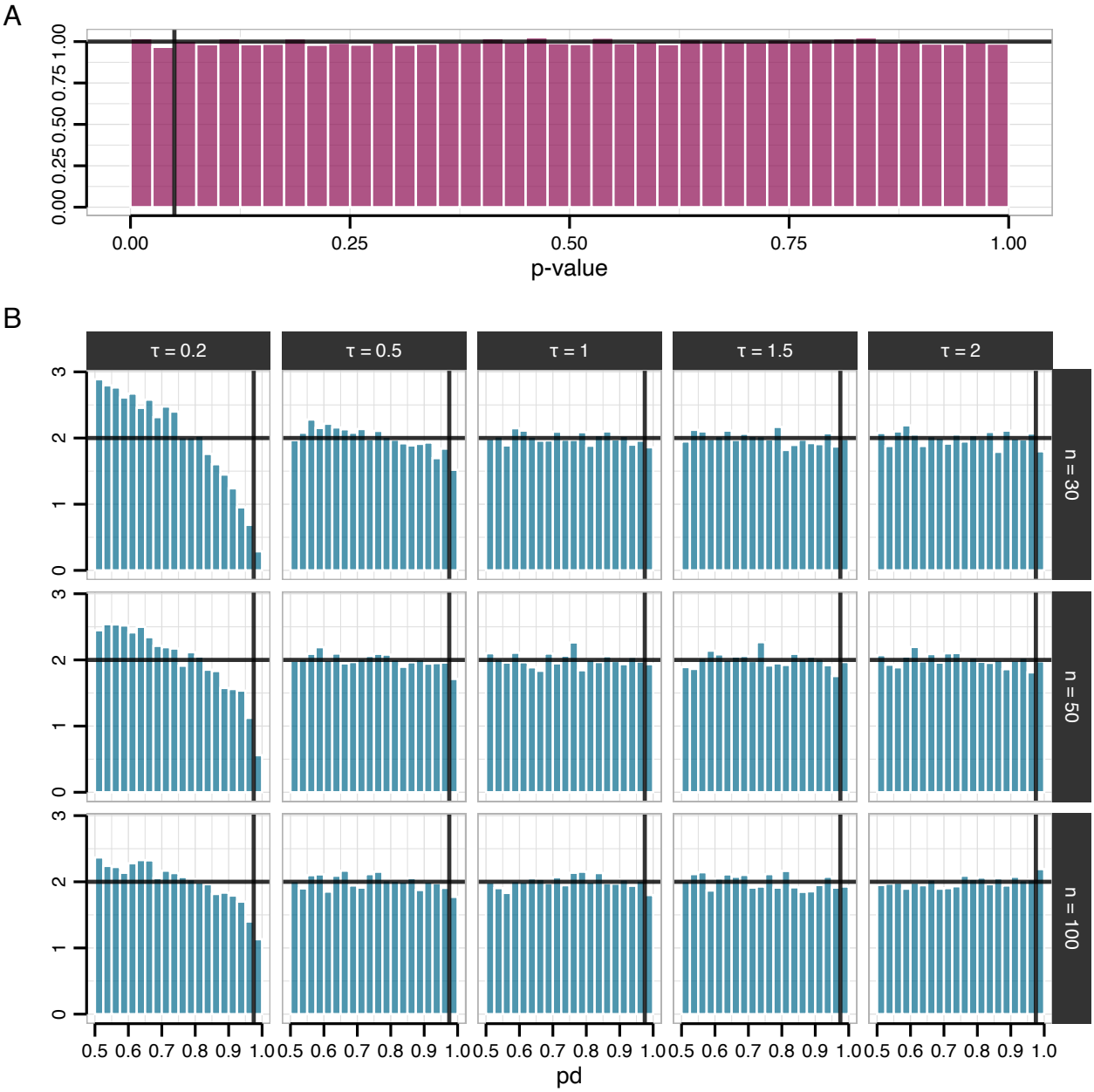
Prior influence

Under the null hypothesis, two-sided p-values are uniformly distributed on $[0, 1]$, and because *pd* can be mapped to the p-value, this implies a corresponding null reference behavior for *pd* on $[0.5, 1]$ in the continuous case. However, because the *pd* is a posterior probability about the sign of θ, its null distribution is not determined by the likelihood alone: it can shift depending on how strongly the prior concentrates probability near 0 (i.e., how readily the prior "expects" near-null effects). Showing this prior dependence at $m = 1$ makes the target of later multiplicity analyses explicit: even before any multiplicity is introduced, a fixed cutoff such as $pd > 0.975$ does not have a universal "null meaning" across priors and sample sizes.

As shown in Figure 3, *pd* exhibits behavior broadly analogous to a p-value when samples are large and the prior places relatively little mass near 0. However, increasing prior skepticism (i.e., concentrating more prior mass around 0) reduces the Type I error rate.

Figure 3

Null reference distributions for p-values and probability of direction. Panel A shows the histogram of two-sided one-sample z-test p-values from 10,000 replications under the global null ($\theta = 0$), with binwidth 0.025. Panel B shows the corresponding empirical distributions of the probability of direction (pd) computed from a conjugate normal posterior, stratified by sample size ($n = 30, 50, 100$; rows) and prior scale ($\theta \sim \mathcal{N}(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$; columns). The vertical dashed line marks the commonly used evidential cutoff (pd = 0.975), which corresponds to a two-sided p-value of 0.05 under the standard conversion $p = 2(1 - pd)$.



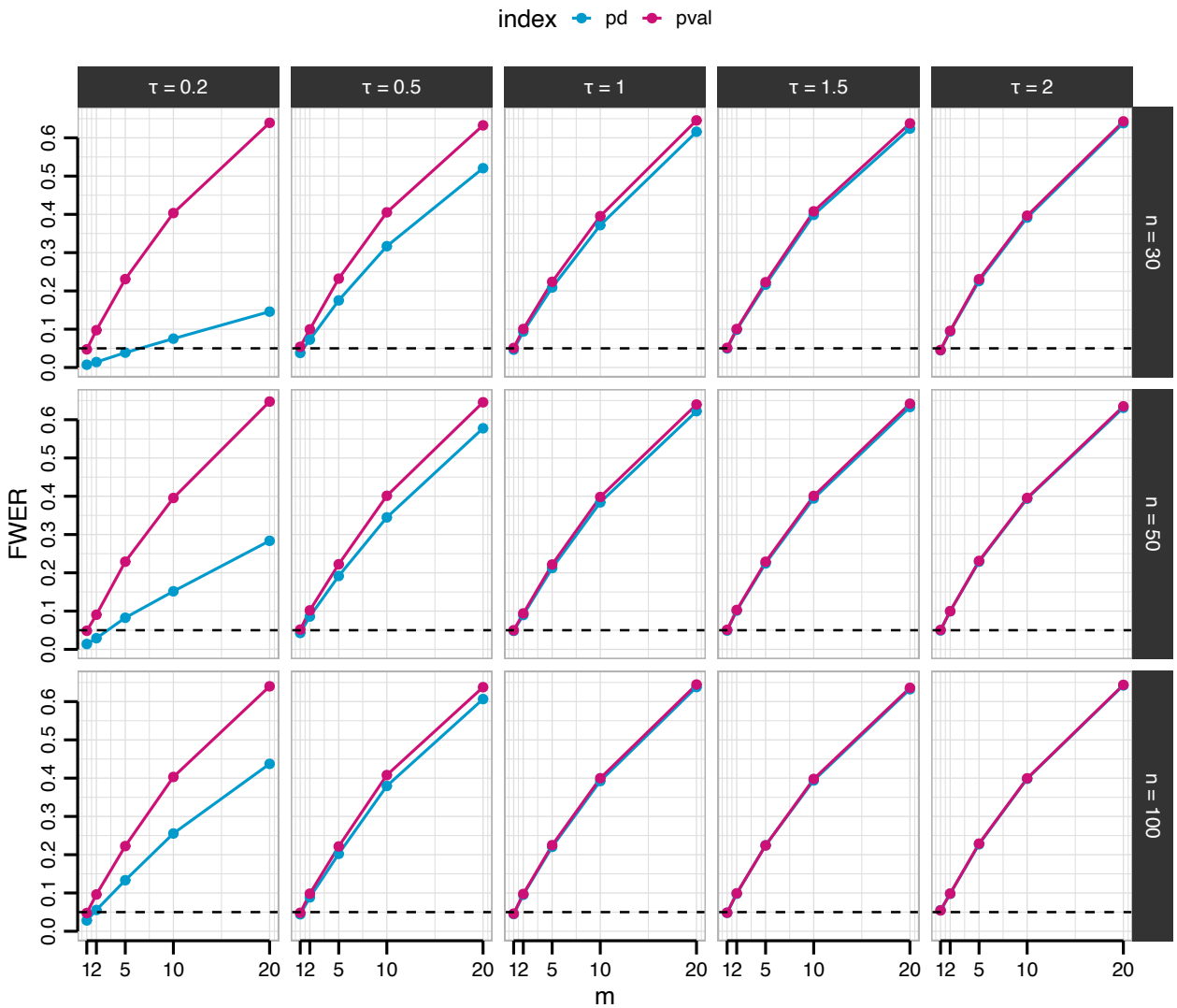
Behavior under multiplicity

Building on this baseline, multiplicity can be introduced as a second, distinct mechanism: even when each individual test has only a small probability of producing an extreme *pd* under H_0, the probability that at least one among $pd_1, \ldots, pd_m$ is extreme increases with the number of tests m. We quantify this familywise error rate (FWER) behavior by estimating *pd* for families ranging from $m = 1$ to $m = 20$.

As expected, *pd* again behaves broadly analogously to the *p*-value when sample sizes are large and the prior places relatively little mass near 0 (Figure 4). By contrast, as the prior becomes more skeptical (i.e., increasingly concentrated around 0), the FWER decreases.

Figure 4

Familywise error rates under the global null for p-values and probability of direction. The figure shows the estimated familywise error rate (FWER), defined as the probability of at least one false positive across m tests, as a function of the number of tests (m), based on 10,000 replications under the global null ($\theta = 0$ for all tests). The pink curve corresponds to declaring at least one discovery when at least one two-sided one-sample z-test p-value < 0.05, whereas the blue curve corresponds to declaring at least one discovery when at least one pd > 0.975 under the conjugate normal Bayesian model. Rows vary the per-test sample size ($n = 30, 50, 100$), and columns vary the prior scale ($\theta \sim \mathcal{N}(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$).



The results of this simulation imply that Type I error can, in principle, be reduced by adopting a more skeptical prior for the parameters within each individual model; however, this is unlikely to be the most appropriate way to address multiplicity. In particular, we think that the prior used to estimate a given parameter should be driven by substantive knowledge about that specific quantity, not by the mere fact that other hypotheses are being tested in parallel [**REF**]. For this reason, the approach proposed below focuses on adjusting between-model prior probabilities (i.e., prior odds across competing hypotheses), rather than modifying the within-model prior for any single parameter [**REF**].

Probability of direction, posterior odds, and prior-odds adjustment

From posterior probabilities to posterior odds

To connect pd to posterior odds, consider the directional hypotheses $H_+ : \theta > 0$ and $H_- : \theta < 0$. These hypotheses partition the parameter space based on the sign of θ. If the prior for θ is continuous and symmetric about 0, then the two directions are equally plausible a priori: $\Pr(H_+) = \Pr(H_-) = 1/2$.

Without loss of generality, assume the posterior favors H_+, so that $pd = \Pr(H_+ \mid y) = \Pr(\theta > 0 \mid y)$ (if the posterior instead favors H_-, we simply relabel the hypotheses), then the complement is given by $1 - pd = \Pr(\theta \leq 0 \mid y)$. The posterior odds comparing these two hypothesis are therefore:

$$\frac{pd}{1 - pd} = \frac{\Pr(H_+ \mid y)}{\Pr(H_{-0} \mid y)}.$$

For a continuous posterior distribution, the probability of any single point (e.g., $\Pr(\theta = 0 \mid y)$) is 0, which means that one could write $\frac{pd}{1-pd} = \frac{\Pr(H_+|y)}{\Pr(H_-|y)}$. However, we retain the notation H_{-0} in the denominator to explicitly indicate that the hypothesis includes the point null value of zero.

Applying Bayes' rule, these posterior odds can be decomposed into a Bayes factor and prior odds:

$$\frac{\Pr(H_+ \mid y)}{\Pr(H_{-0} \mid y)} = \mathrm{BF}_{+/-0} \times \frac{\Pr(H_+)}{\Pr(H_{-0})},$$

where BF+/−0 quantifies the relative likelihood of the data under $H+$ versus H_{-0}. Under prior

symmetry, $\frac{\Pr(H_+)}{\Pr(H_{-0})} \approx 1$, which yields $\frac{pd}{1-pd} = \mathrm{BF}_{+/-0}$ (Casella & Berger, 1987; Marsman & Wagenmakers, 2017). Thus, the odds $\frac{pd}{(1-pd)}$ computed from posterior draws can be interpreted directly as evidence favoring $\theta > 0$ over $\theta \leq 0$ (or $\theta < 0$ over $\theta \geq 0$).

Multiplicity adjustment via post-hoc prior-odds

When assessing a single directional claim under a symmetric prior centered at the null, $\frac{\Pr(H_+)}{\Pr(H_{-0})} = 1$ encoding prior indifference. However, when m hypotheses are tested simultaneously, assigning prior probability $\frac{1}{2}$ to each directional claim implies a joint prior that rapidly concentrates probability on the event that at least one effect departs from 0. For instance, with $m = 10$ the implied prior probability that all effects are not in the favored direction (i.e., in the opposite direction with zero included) is $0.5^{10} \approx 0.001$.

In this scenario, a principled way to address multiplicity is to decrease the prior odds assigned to any particular directional claim as a function of m, while leaving the the Bayes factor component unchanged. Because the analysis is symmetric in the sign of θ and pd is defined as the posterior probability of the dominant direction, the same adjustment can be applied regardless of whether the posterior favors $\theta > 0$ or $\theta < 0$.

Concretely, let $\Pr(H_{1_m})$ and $\Pr(H_{0_m})$ denote the multiplicity-adjusted prior probabilities assigned to the favored direction and its complement (i.e., opposite direction with zero included), respectively (with $\Pr(H_{0_m}) + \Pr(H_{1_m}) = 1$). The adjusted posterior odds are then:

$$\frac{pd_{\text{adj}}}{1 - pd_{\text{adj}}} = \frac{pd}{1 - pd} \times \frac{\Pr(H_{1_m})}{\Pr(H_{0_m})},$$

so that, equivalently,

$$pd_{\text{adj}} = \frac{pd \cdot \Pr(H_{1_m})}{pd \cdot \Pr(H_{1_m}) + (1 - pd) \cdot \Pr(H_{0_m})}.$$

Thus, the multiplicity adjustment acts through the prior-odds factor $\Pr(H_{1_m})/\Pr(H_{0_m})$.

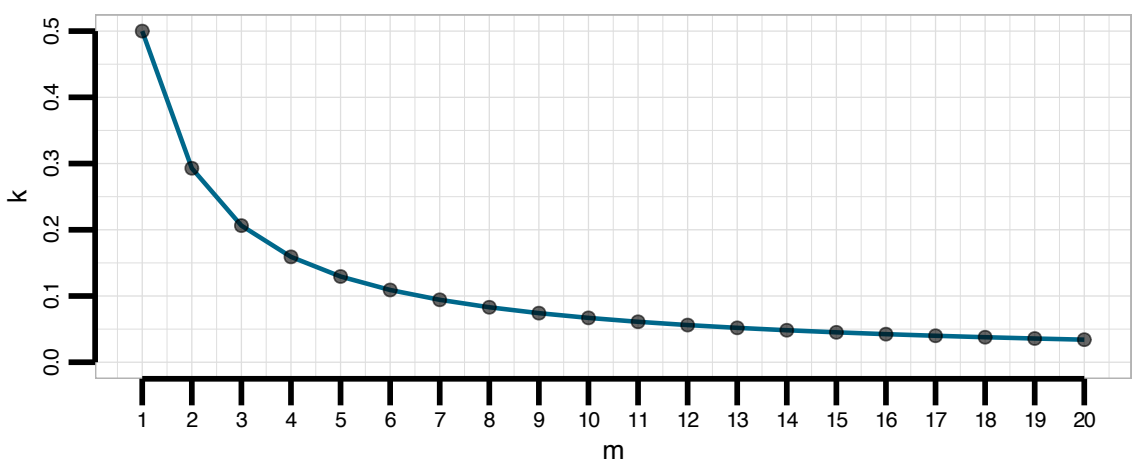
Choosing the prior-odds

Following the global-null formulation of Jeffreys (1938; see also Westfall et al., 1997), we introduce a global null event Q and its complement $\neg Q$, and assign equal prior probability

$\Pr(Q) = \Pr(\neg Q) = 1/2$. Define, for each $j = 1, \ldots, m$, the "favored-direction" alternative $H_{1,j}$ as the directional hypothesis corresponding to the posterior-dominant sign (i.e., the direction used in $pd_j = \Pr(H_{1,j} \mid y)$), and let $H_{0,j}$ denote its complement (opposite direction with 0 included). We then set $Q = \bigcap_{j=1}^{m} H_{0,j}$ (no parameter lies in its favored direction) and $\neg Q$ as its complement (at least one parameter lies in its favored direction). Under independence and a common per-hypothesis prior probability $k = \Pr(H_{1,j})$ for all j, this implies $\Pr(Q) = (1-k)^m$, so $(1-k)^m = 0.5$ and therefore $k = 1 - 0.5^{1/m}$. As m increases, the per-hypothesis prior probability $k = \Pr(H_{1,j})$ decreases (Figure Figure 5), encoding increasing prior skepticism about any particular deviation from the null; correspondingly, $\Pr(H_{0,j})$ increases.

Figure 5

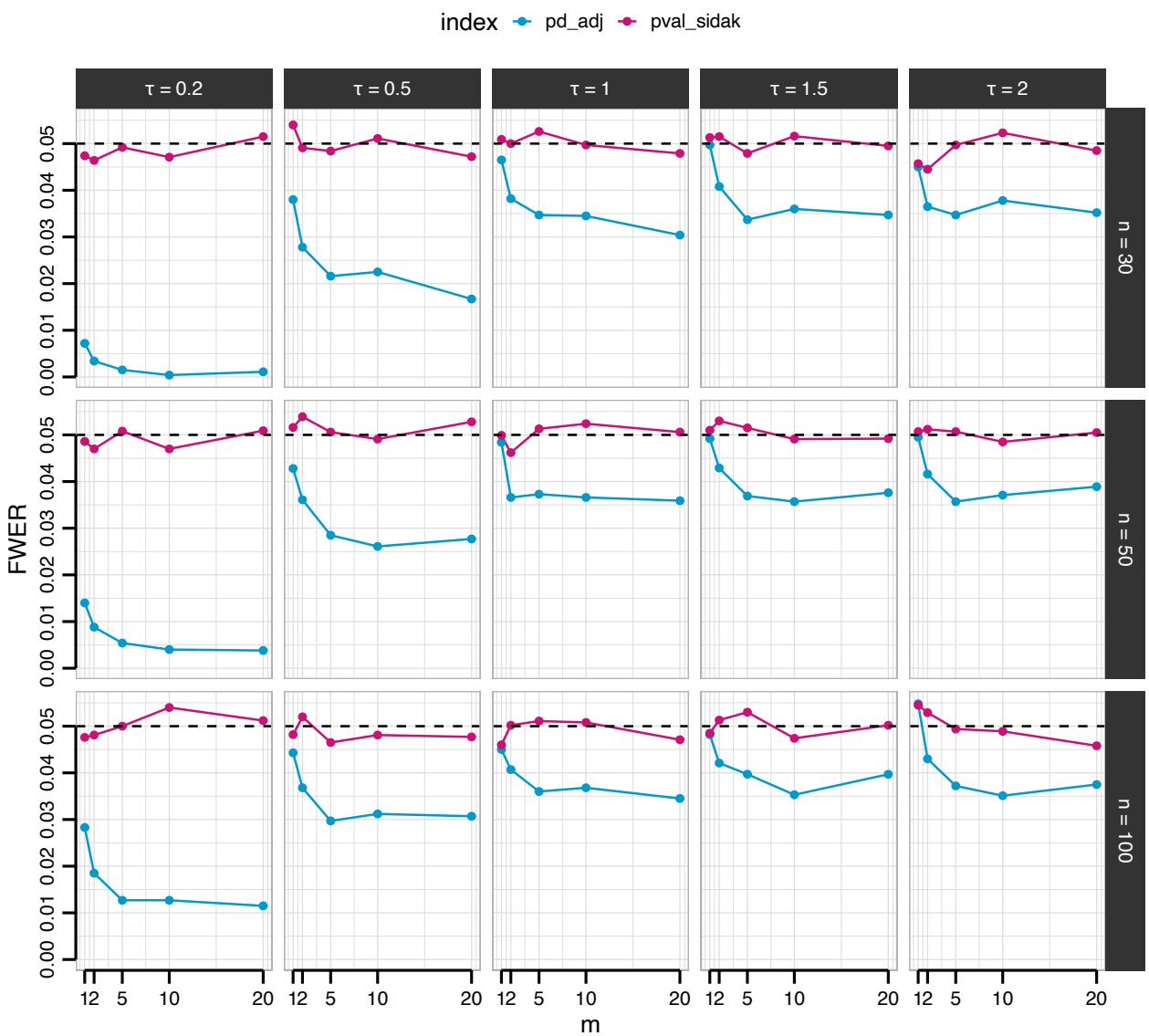
Per-hypothesis prior probability k s a function of the total number of hypotheses m, with the overall prior probability of at least one effect fixed at $q = 0.5$. *The per-hypothesis prior* $\Pr(H_{1,j})$ *is given by* $k = 1 - 0.5^{1/m}$.



In this simulation, we compare this method with the Šidák p-value adjustment ($p_{adj} = 1 - (1-p)^m$), which gives exact frequentist family-wise error rate control under independence. While grounded in Bayesian prior-odds logic, the procedure exhibits behavior similar to frequentist multiplicity correction, though it tends to be more conservative even when n and τ are large (Figure 6).

Figure 6

Familywise error rates under the global null for adjusted p-values and probability of direction. The figure shows the estimated familywise error rate (FWER), defined as the probability of at least one false positive across m tests, as a function of the number of tests (m), based on 10,000 replications under the global null ($\theta = 0$ for all tests). The red curve corresponds to declaring at least one discovery when at least one two-sided one-sample z-test p-value < 0.05, whereas the blue curve corresponds to declaring at least one discovery when at least one pd > 0.975 under the conjugate normal Bayesian model. Rows vary the per-test sample size ($n = 30, 50, 100$), and columns vary the prior scale ($\theta \sim \mathcal{N}(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$).



Having linked the multiplicity adjustment to the global null event Q, the same construction yields a more general calibration directly in terms of $\Pr(Q)$. Specifically, instead of fixing $\Pr(Q) = 0.5$, researchers may encode background knowledge by choosing $q = \Pr(Q)$, the prior probability that all m hypotheses are null. Under a priori independence and a common per-hypothesis prior probability $k = \Pr(H_{1,j})$ for the favored-direction alternative, $\Pr(Q) = (1-k)^m$, so choosing q implies

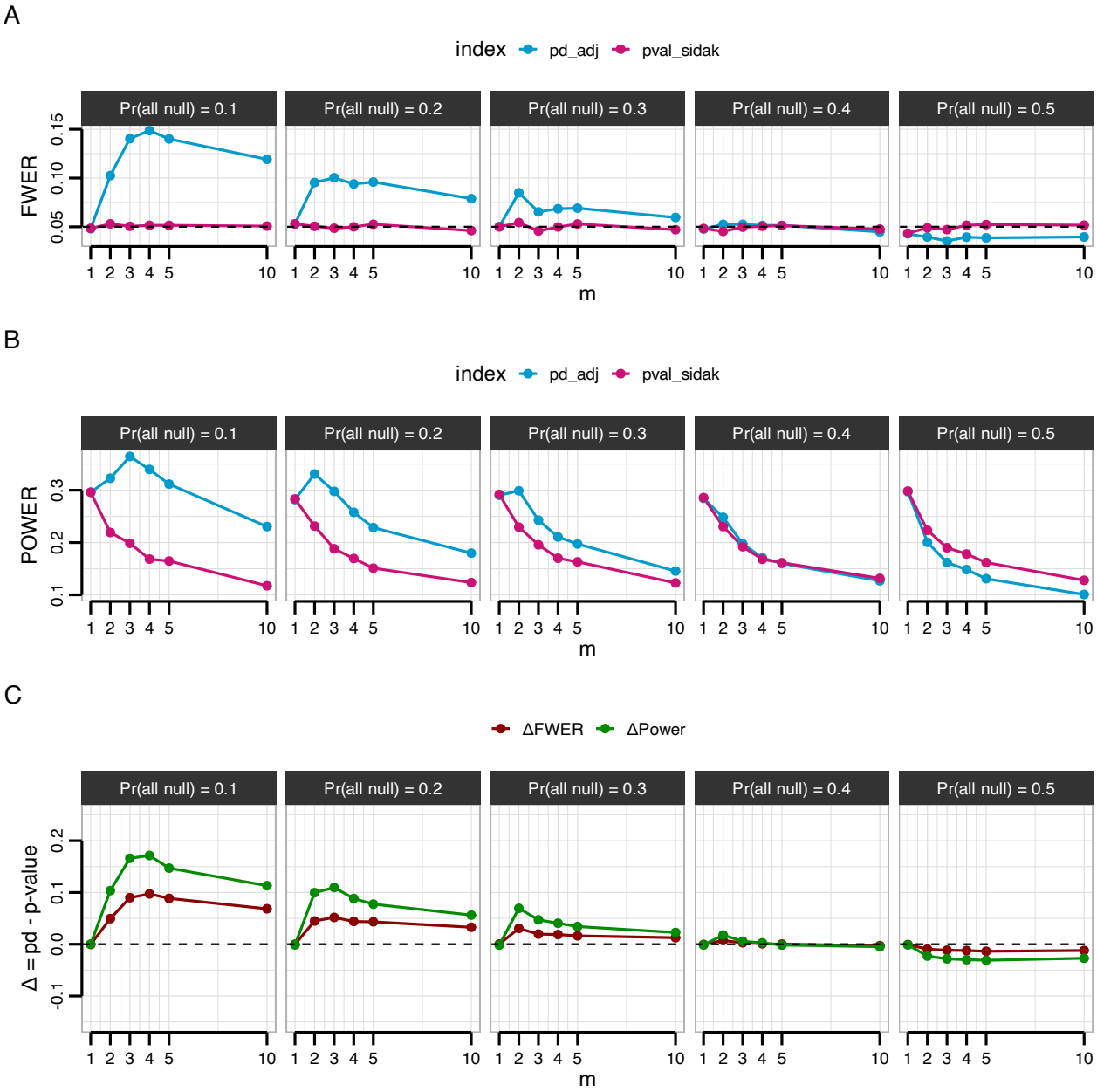
$$(1-k)^m = q \quad \Rightarrow \quad k = 1 - q^{1/m}.$$

Prior knowledge guides q: when few (if any) non-null effects are expected, q should be large; when effects are well established, q can be smaller. For small m and sufficiently small q, however, the calibration can become anti-conservative: $\Pr(H_{0,j}) = 1 - k = q^{1/m}$ may fall below $1/2$, which implies $k > 1/2$ and therefore increases rather than decreases the odds for $H_{1,j}$. Because the goal of the adjustment is to guard against inflated false positives, it should never move posterior probabilities in this direction; accordingly, a simple safeguard is to leave posterior probabilities unchanged whenever $\Pr(H_{0,j}) < 1/2$ (equivalently, whenever $q^{1/m} < 1/2$).

Once q is fixed, the prior odds assigned to each individual hypothesis become a deterministic function of the family size m: larger families imply smaller k, and consequently a higher prior probability on the null for each test. To illustrate this dependence, we consider two data-generating regimes: a global-null scenario in which $\theta_1 = \cdots = \theta_m = 0$, and an alternative scenario in which all but one effect are null ($\theta_1 = \cdots = \theta_{m-1} = 0$, $\theta_m = 0.2$). We then repeat the previous simulation procedure with $\tau = 2$ and $n = 50$, i.e. a setting in which *pd* and *p*-value behavior is broadly comparable, while varying only the number of tests $m \in \{1, 2, 3, 4, 5, 10\}$ and the prior parameter $q \in \{0.50, 0.40, 0.30, 0.20, 0.10\}$.

Figure 7

Familywise error rate (Panel A), power (Panel B), and difference between method (Panel C) for adjusted p*-values and probability of direction as a function of the number of tests (m) and prior* Pr(*all* $\theta = 0$). *The Šidák-adjusted* p*-value curve (pink) declares at least one discovery when any two-sided one-sample* z*-test satisfies* $p < 0.05$*, whereas the adjusted* pd *curve (blue) declares at least one discovery when any* $pd > 0.975$ *under the conjugate normal Bayesian model. Panel C shows the difference (adjusted pd minus Šidák-adjusted p-value) for FWER (under the global null) and power (under the global alternative).*



Across all values of m, Šidák-adjusted p-values maintain the family-wise error rate (FWER; Figure 7, Panel A) near the nominal level, but this control comes at the expense of diminished power (Figure 7, Panel B). By contrast, the adjusted *pd* procedure exhibits FWER and power trajectories that depend on the prior parameter q. For smaller values of q ($q \in \{0.30, 0.20, 0.10\}$), the tradeoff is favorable: the increase in power is larger than the corresponding increase in FWER, implying gains relative to the loss of error control (Figure 7, Panel C). By contrast, $q = 0.4$ yields FWER and power curves that closely mirror the Šidák adjustment across the full range of m, suggesting that this calibration allows the Bayesian adjustment to approximate classical strong FWER control (Figure 7).

Application

Discussion

Our simulations demonstrate that extreme **pd** values arise by chance with non-negligible frequency, and this problem compounds as researchers evaluate more parameters. To address this, the multiplicity adjustment proposed here accommodates researcher-specific prior information, which may be informed by previous studies, measurement reliability, or structured expert elicitation [**REFs**].

An advantage of this procedure is its applicability to **pd** values computed from posterior distributions obtained under different models. While the most coherent approach analyzes all relevant data within a single joint model that explicitly represents dependence [**REFs**], this is not always feasible in practice, and researchers often report pd across multiple analyses within the same manuscript [**REFs**]. The procedure outlined here provides a flexible toolkit for such scenarios. Additionally, the approach naturally extends beyond point-null testing by evaluating directional certainty relative to a smallest effect size of interest, shifting emphasis from non-zero effects to substantively meaningful thresholds (Lakens et al., 2018).

The procedure does assume independence between hypotheses when allocating global null probability across m tests, treating each as providing independent evidence. When hypotheses are correlated through shared parameters, overlapping data structures, or measurement dependencies,

this independence-based allocation overestimates the multiplicity penalty, yielding overly conservative adjustments [**REFs**]. A practical solution is to replace the nominal number of tests m with an effective number m_{eff} that reflects the dependence structure. This can be achieved by computing a correlation matrix R that captures correlation among tests (e.g., posterior distributions from the same model) and mapping R to m_{eff} via eigenvalue-based methods (Cheverud, 2001; Del Giudice, 2021; Li & Ji, 2005). The resulting m_{eff} will generally depend on how R is defined and estimated, so it should be treated as a sensitivity parameter rather than a uniquely determined quantity.

For applied reporting, we recommend a two-tier strategy: first, report the unadjusted *pd* for each parameter to maintain transparency; second, report at least one multiplicity-adjusted measure using prior-odds correction. As a baseline, researchers should present the conservative adjustment with $\Pr(\text{all } \theta = 0) = 0.4$, which represents a skeptical but defensible stance as it produces adjustments analogous to the Šidák correction for p-values. When domain knowledge justifies alternative priors, report both conservative and informed adjustments to demonstrate robustness across different assumptions.

Regardless of its limitations, we believe this procedure provides a principled approach to Bayesian multiplicity adjustment with meaningful implications for both interpretability and replicability of research findings.

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